

1. Summary of the submission

Mark	25%
Group size	Already created groups.
Deadline	May 19, 2026, at 23:59

2. Instructions

This assignment requires the application of various time series forecasting techniques that we have discussed in class, utilizing the previously selected dataset. You should submit a Jupyter Notebook or Google Colab file that includes comprehensive documentation and is fully executable. Ensure that the dataset is publicly accessible via Google Drive. Also, maintain all console outputs within the notebook.

Required Sections

1. Time series analysis:

- Visualize the time series to illustrate its length and structure. Apply decomposition techniques to determine the presence of trends, seasonality, and residuals.

2. ARMA.

- Results obtained using this modeling procedure. If it is not possible to apply it, justify why.

3. ARIMA (General modeling procedure)

- Results obtained using this modeling procedure. Test at least two different combinations of p (order of autoregression) and q (order of moving average).
- If seasonality is present, consider employing a naïve approach to manage this characteristic using ARIMA.

4. SARIMA (General modeling procedure adapted for SARIMA)

- Results obtained using this modeling procedure.
- If your time series does not have any seasonality, you should justify why.

5. AutoArima or SARIMAX(if you have exogenous variables)

- Results obtained using this modeling procedure.
- With and without seasonality.

6. Transformer based architecture

- Use one transformers-based architecture for time series forecasting.

7. Results discussion

- Present the results obtained from your models, applying the same metric across all the models. Establish some baselines (last known value, mean, naïve seasonal...).
- Use a table to compare all the results
- Highlight any interesting findings observed during your analysis.

8. Lessons Learned

- Identify key challenges you faced during the project and how you overcame them.

- Share any insights gained from working with time series and from selecting specific parameters (window, horizon, range of values for q and p ...).